



Nutrients as signals that spatially organize plant–microbe interactions

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Plant–microbe interactions are spatially organized processes. Increasing evidence suggests that nutrients play a critical role in this organization by not only acting as resources but also as spatial signals. In plants, nutrient perception and signaling are spatially patterned across tissues, while in microbes, local nutrient conditions guide colonization, metabolism, and function. These responses create heterogeneous interaction niches at the plant–microbe interface, but how they are coordinated across individual cells remains unclear. Recent advances in single-cell, spatial omics and nutrient mapping approaches now enable these processes to be resolved at cellular resolution. Here, we bring together current understanding of nutrient-guided plant and microbial responses and highlight emerging technologies that allow their integration. We then introduce a cell-state framework to interpret plant–microbe interactions as emergent properties of spatially coupled cellular responses linked through local chemical environments. This perspective provides a foundation for a more mechanistic and predictive understanding of plant–microbe interactions.

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and longitudinal axes. Distinct microbiome compositions occur across root-associated compartments (rhizosphere, rhizoplane, and endosphere) [1,4–6] and across developmental zones [7*]. This raises the question: how are such spatially distinct microbial niches generated and maintained within a single host?

One possible explanation is that nutrients act as key drivers of spatial heterogeneity in plant–microbe interactions. Soil nutrient availability is inherently heterogeneous, shaped by physicochemical properties and fluctuating biotic and abiotic conditions. Plants and microbes have evolved strategies to adapt to these dynamic landscapes, influencing their interactions. In plants, nutrient status modulates developmental and immune programs, influencing microbial colonization. In microbes, nutrient cues reshape metabolic states and behaviors, altering host engagement. Microbial communities facilitate plant nutrient acquisition, while plants shape microbial nutrient capture and cycling through carbon-rich exudates and other rhizodeposits [8]. Thus, nutrients function not only as resources but also as spatial signals that structure local interaction niches. Yet how nutrient-mediated interactions are organized at cellular resolution—how individual plant and microbial cells process nutrient cues and collectively shape their interactions—remains poorly understood.

Recent advances in single-cell and spatial omics now provide an opportunity to address these gaps by enabling high-resolution mapping of cellular identities and spatial organization in plant and microbial systems [9]. In this review, we first examine how nutrients act as spatial signals shaping plant and microbial decisions, focusing on root–microbe interactions. We then introduce the emerging tools that can resolve them simultaneously, and finally propose a cell-state framework for understanding plant–microbe interactions at cellular resolution within shared nutrient microenvironments.

Introduction

Plants interact with diverse microbial communities, spanning relationships from mutualism to pathogenicity [1–3]. Roots serve as the primary interface with soil and host particularly diverse microbiota. These associations are not random but are spatially organized along radial

Nutrient availability shapes spatially heterogeneous plant cellular responses

To navigate heterogeneous soil environments, plant nutrient perception, signaling, and transport are spatially compartmentalized within tissues. Transporters and receptors show cell type- and zone-specific